



Introduction to CFD (4RC30)

Outlook

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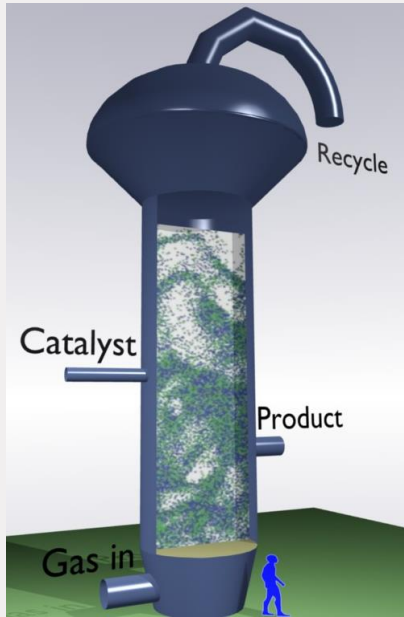
Outline

Content of this lecture

- Multiphase flow modelling
 - Multiscale modelling strategy
 - Examples
- Commercial CFD packages
 - Pro's and con's
 - Examples

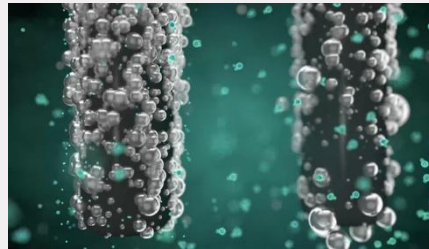
Wrap up

Multiphase flows: Multiscale



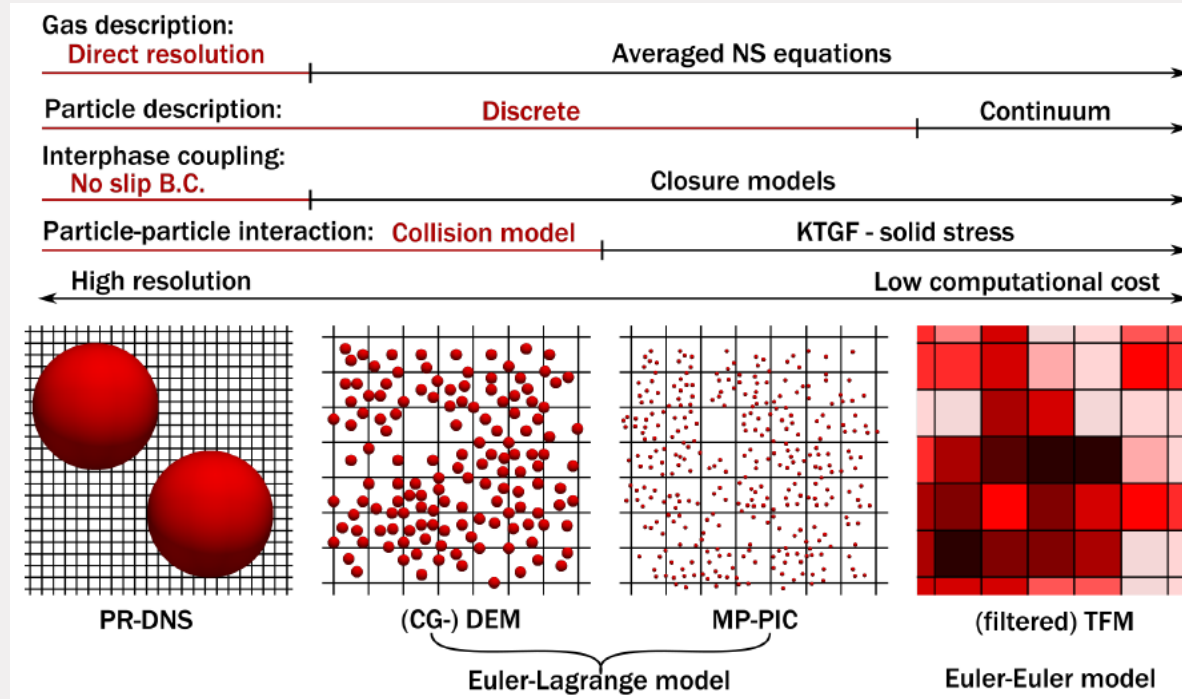
Fluidized bed reactor:

- Fluid Catalytic Cracking
- Granulation
- Combustion



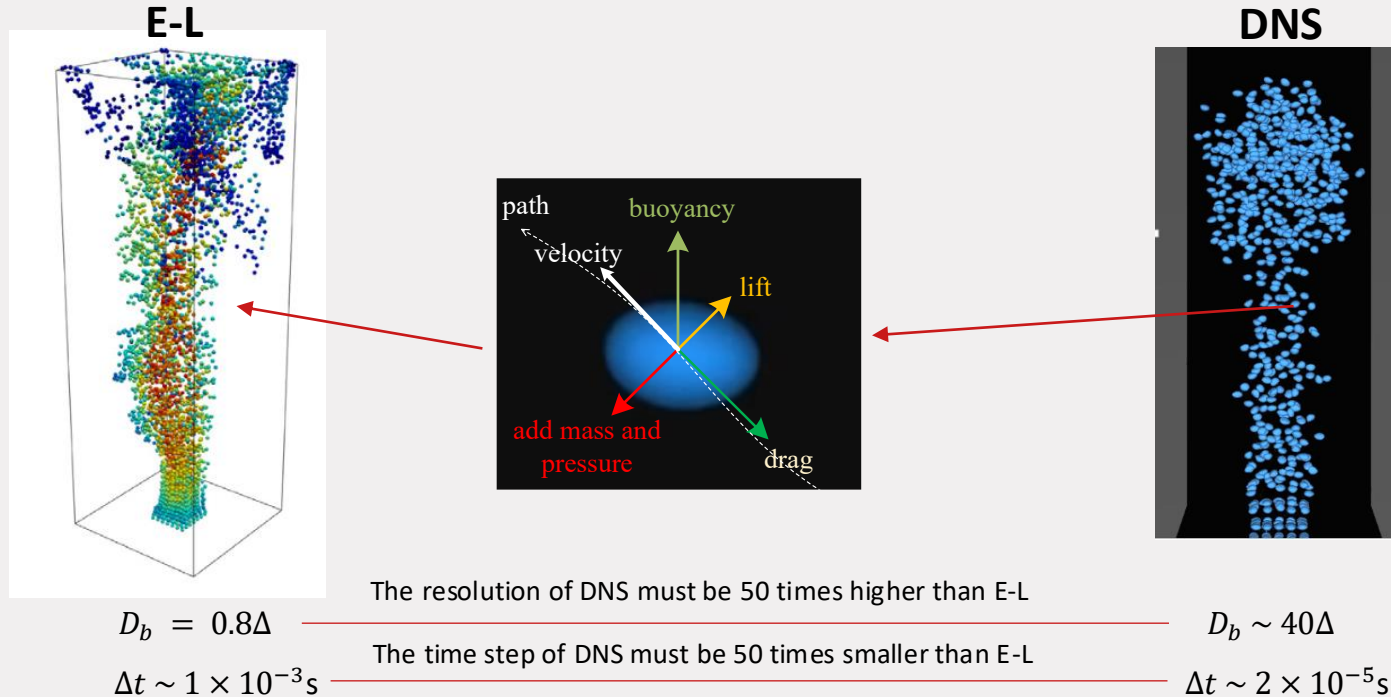
Global electrolyzer capacity could reach 134 GW by 2030

Multiscale Modelling



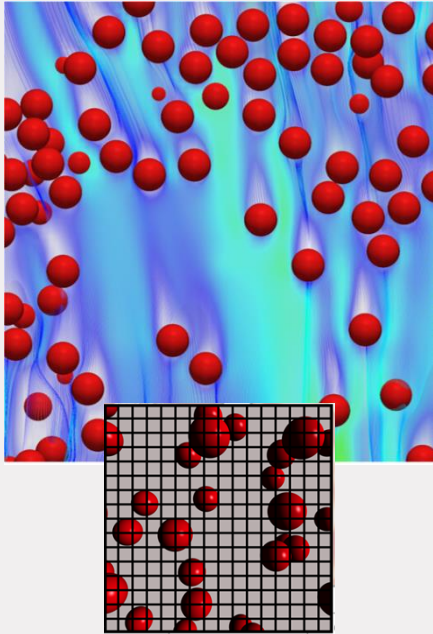
larger geometry, less details

Multiscale Modelling



The computation of DNS is about 6.25 million times larger than the cost of E-L !!!

Why Direct Numerical Simulations?



DNS Approach and Role

- Fully resolve all continuum scales without using sub-grid models or major assumptions
- Clear definition of all conditions (initial, boundary and forcing) and the production of data for every single variable
- DNS is (currently) limited to small domains due to large computational requirements. Device-scale simulations are out of reach.
- A tool for fundamental studies of the small scale physics of multiphase flows
- A tool for the development and validation of reduced model descriptions used in meso- and macro- scale CFD simulations

DNS: drag force on particles

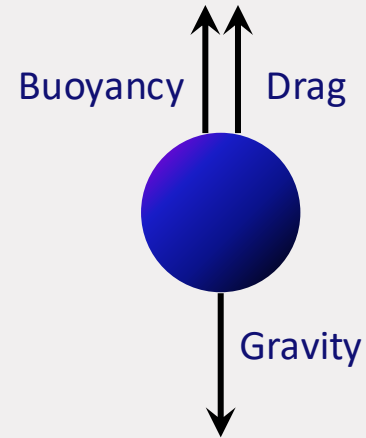
Definition of drag coefficient C_D :

A fluid with (relative) approach velocity $\mathbf{v}_{rel}$ exerts a force $\mathbf{F}$ on the object:

$$\mathbf{F}_D = C_D A \left(\frac{1}{2} \rho \mathbf{v}_{rel}^2 \right) \quad [\text{N}]$$

Drag coefficient $\nearrow$
Projected area $\uparrow$
Kinetic energy per unit volume $\nearrow$

where $C_D = f(\text{geometry}, \text{Re}) \quad [-]$



DNS: drag force on particles

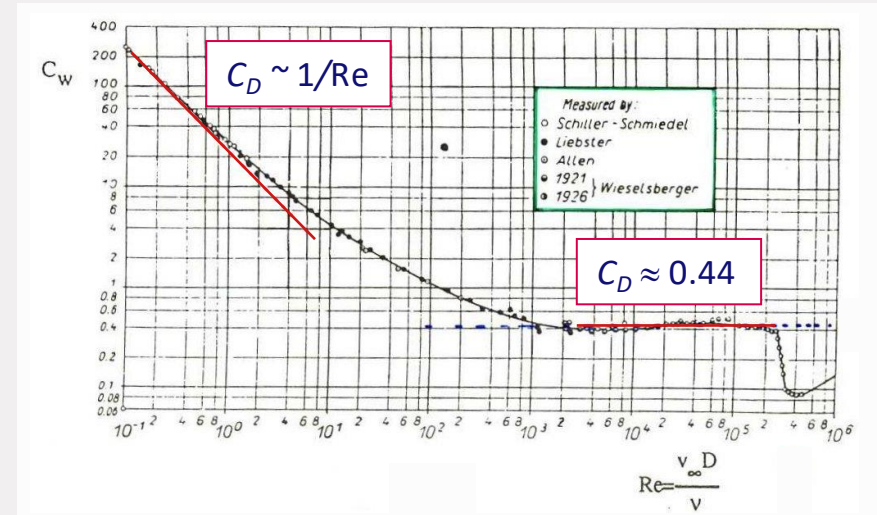
Drag coefficient on a spherical particle

- Stokes flow:

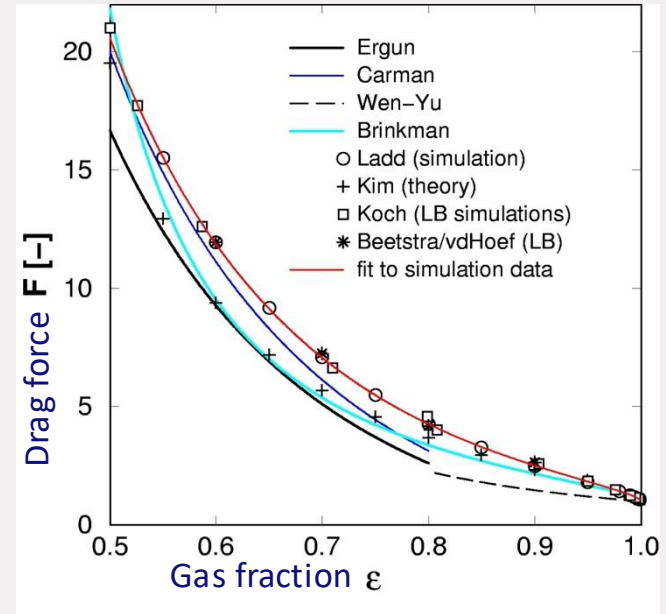
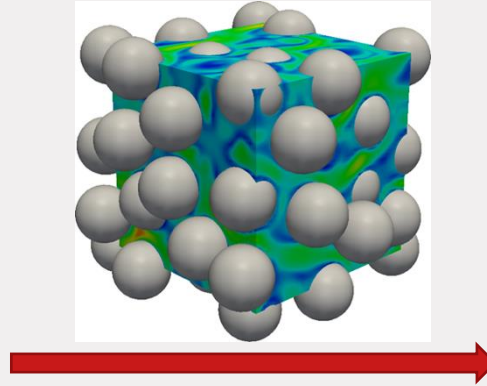
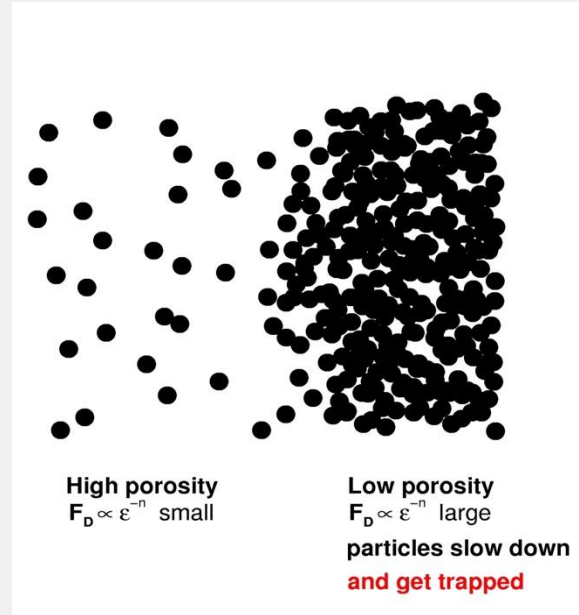
$$C_D = \frac{24}{\text{Re}_p}$$

- Correlation valid for finite Reynolds number

$$C_D = \begin{cases} \frac{24}{\text{Re}_p} (1 + 0.15 \text{Re}_p^{0.687}) & \text{Re}_p < 1000 \\ 0.44 & \text{Re}_p > 1000 \end{cases}$$

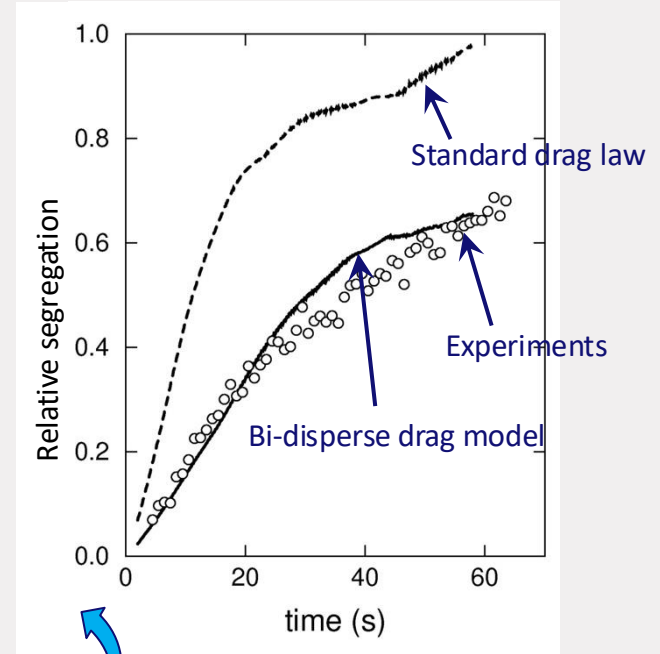
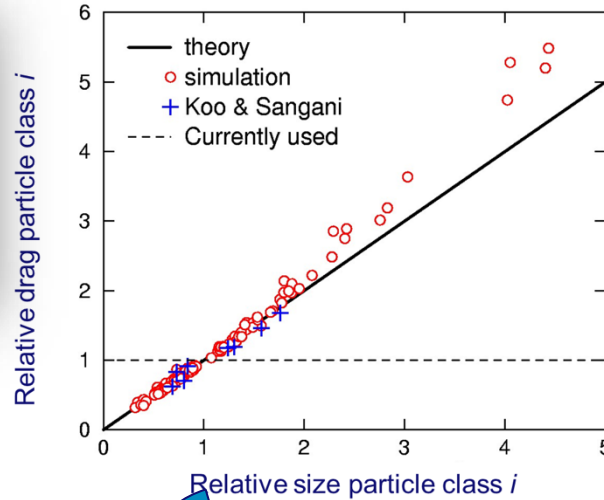


DNS: drag force on particles



$$F_d = F(\varepsilon, Re)$$

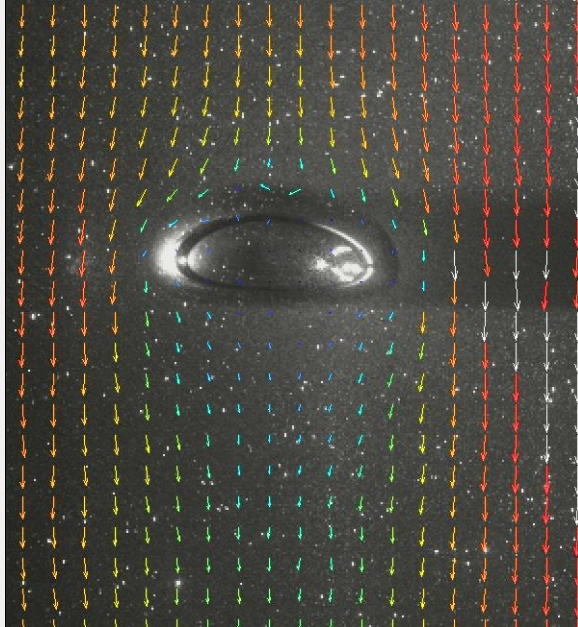
DNS: drag force on particles



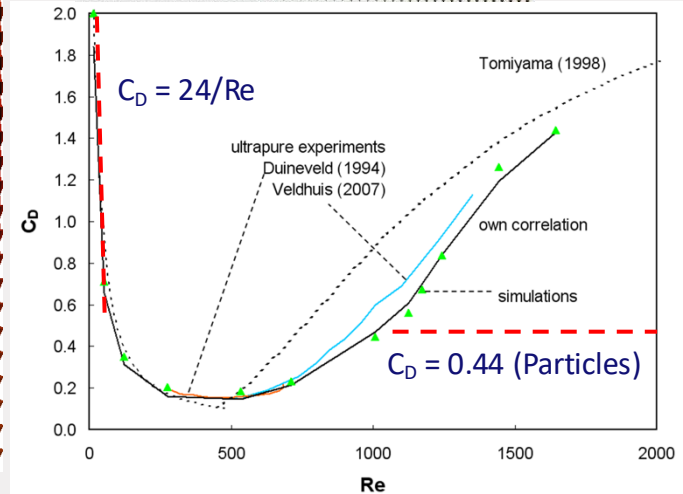
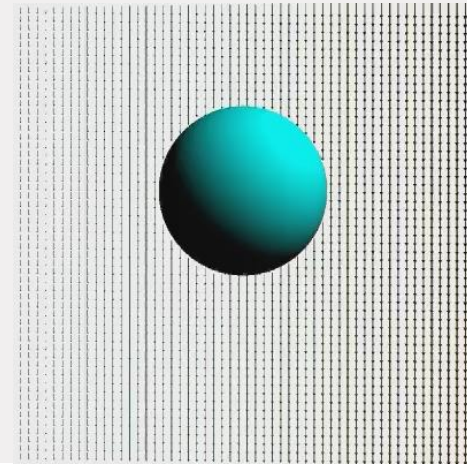
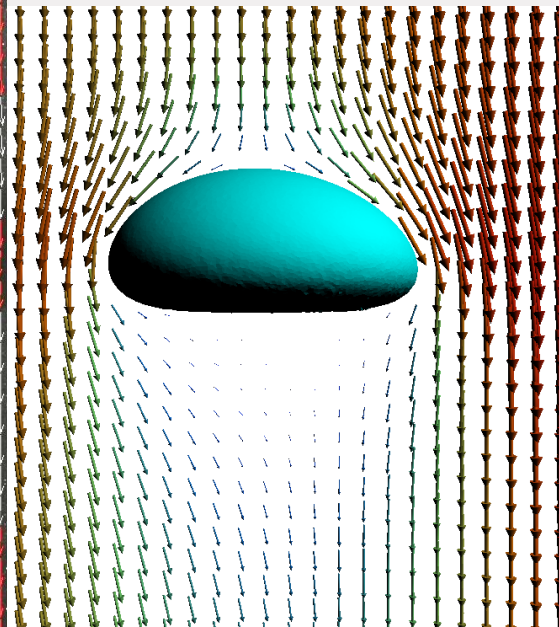
$$F_i = F(\varepsilon_s, \text{Re}) \quad \text{with} \quad y_i = \frac{d_i}{\langle d \rangle}$$

DNS: drag force on bubbles

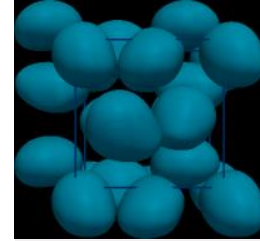
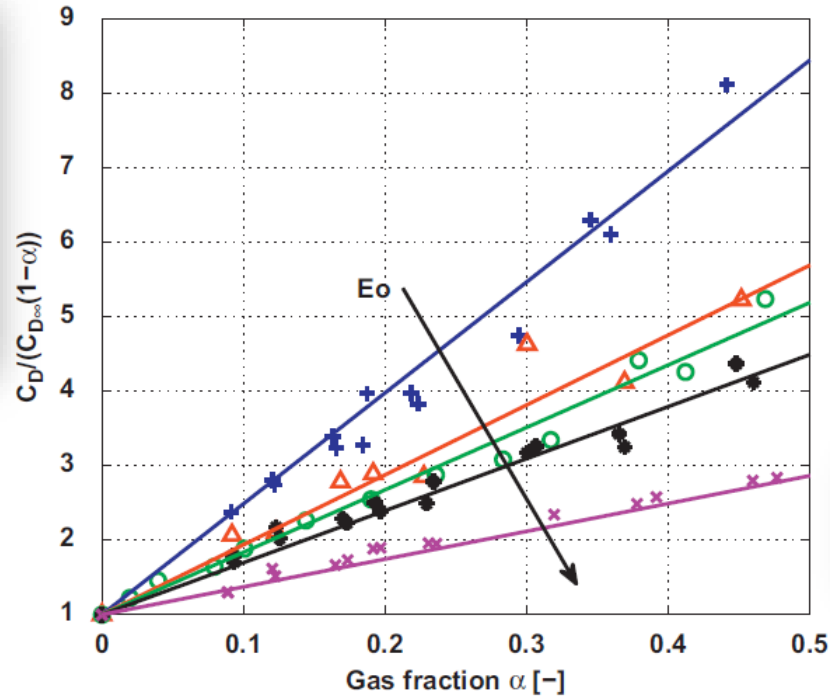
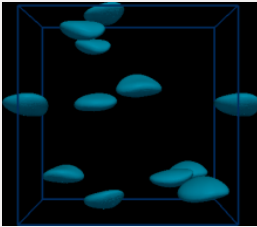
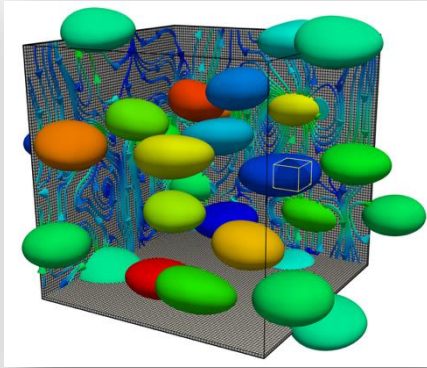
Experiment



Simulation



DNS: drag force on bubbles



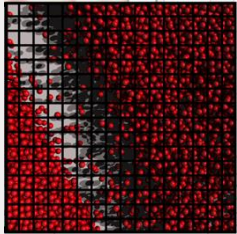
$$\frac{C_D}{C_{D,\infty}(1-\alpha)} = 1 + \frac{18}{Eo} \alpha$$

DPM

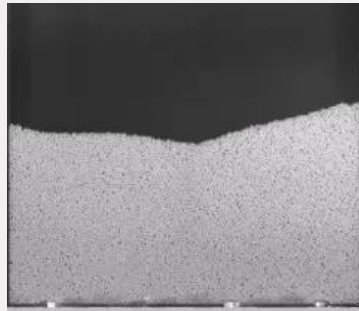
Gas phase: Navier-Stokes eq.

Solid phase: Newton's 2nd law

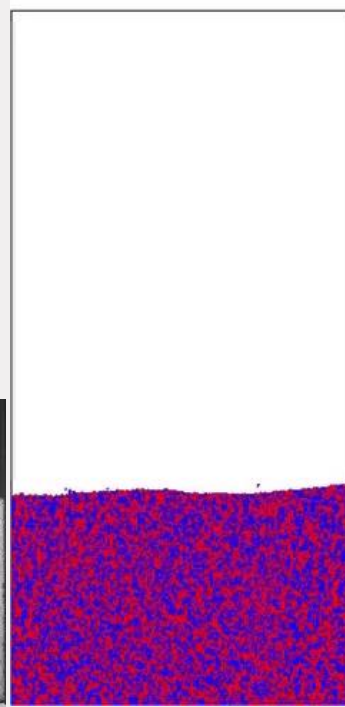
Require (DNS-based) drag correlation



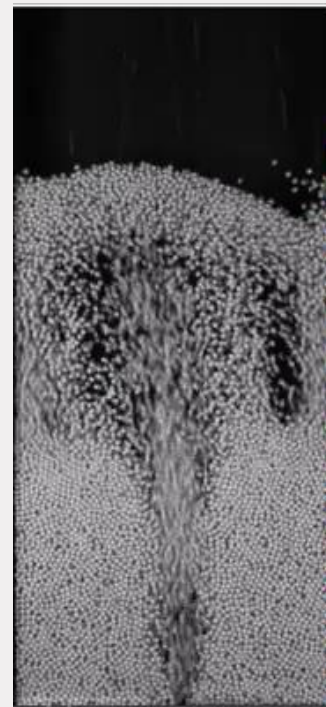
Experiment



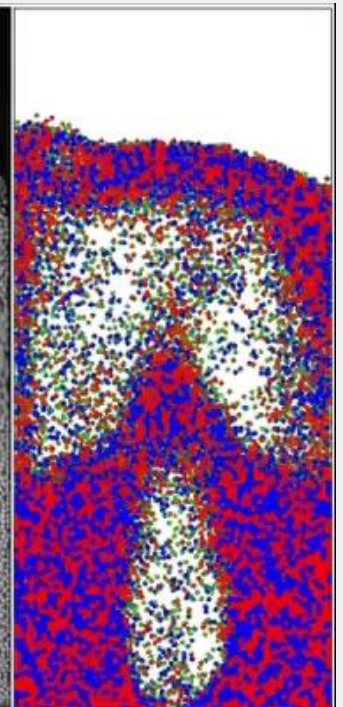
Simulation



Experiment

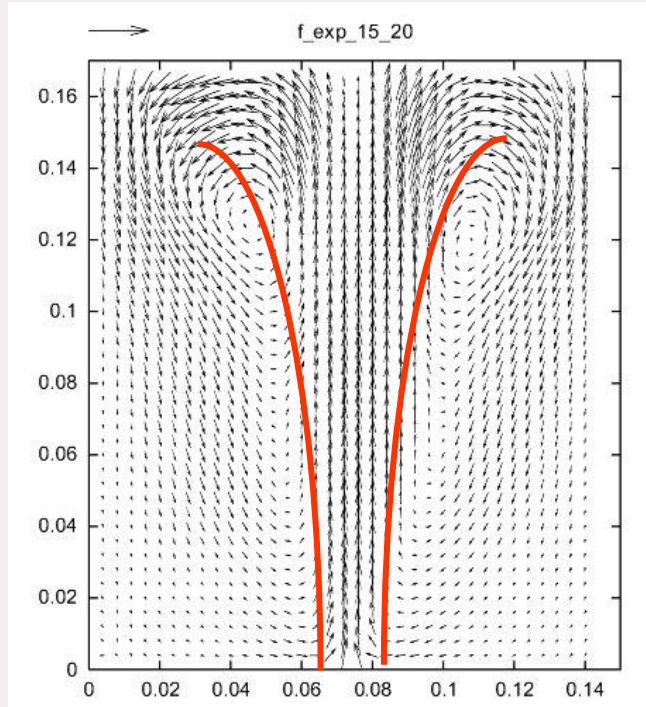


Simulation

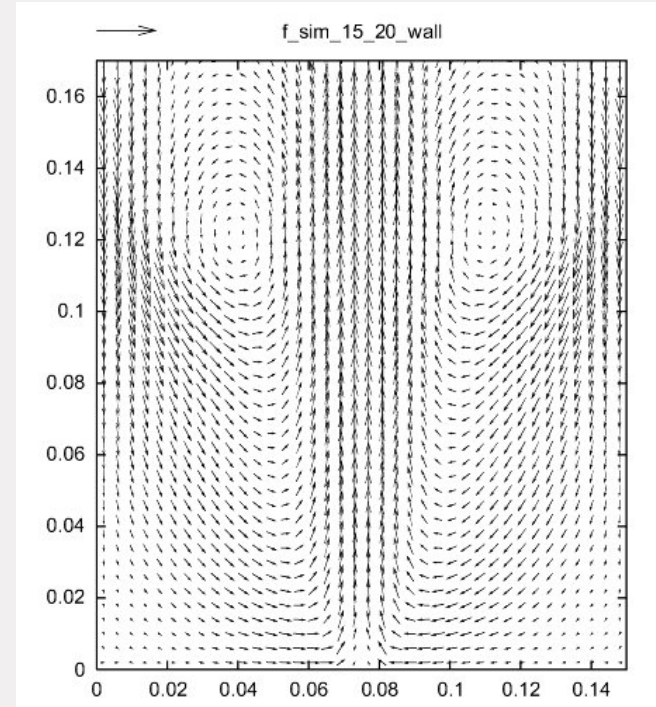


DPM: steady spouted bed

Experiment



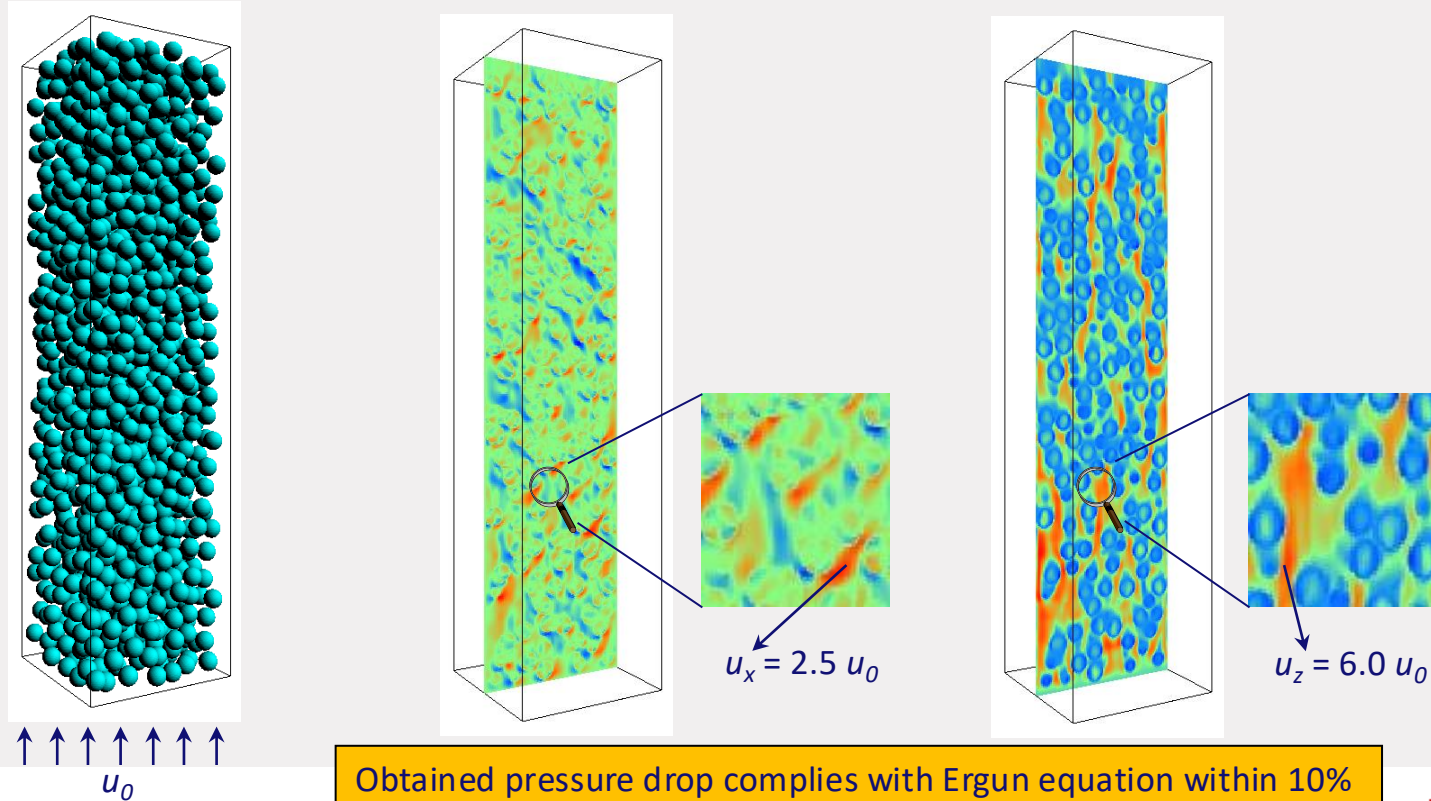
Simulation



DPM: effect of operation pressure



DPM: packed bed reactor



Obtained pressure drop complies with Ergun equation within 10%

Two-Fluid Model (TFM)

Gas phase and solid phase: Volume-averaged Navier-Stokes eqs.

Require drag correlation for gas-solid interaction

Require model to describe particle-particle interactions

- Bubble size in a bubbling fluidized bed:

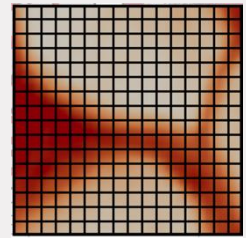
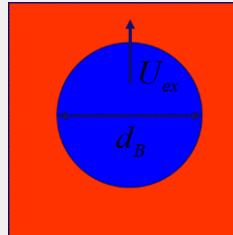
$$\frac{dV_B}{dt} = Q - U_{ex} A_B \quad V_B = \frac{\pi}{6} d_B^3 \quad A_B = \pi d_B^2$$

- Two simple models:

$$U_{ex} = 0 \quad U_{ex} = U_{mf}$$

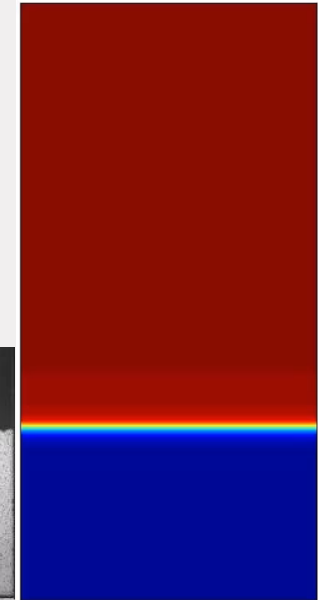
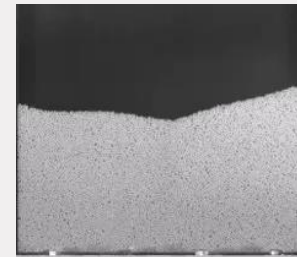
- ... and a multiphase CFD model

Kinetic theory of granular flow (KTGF)



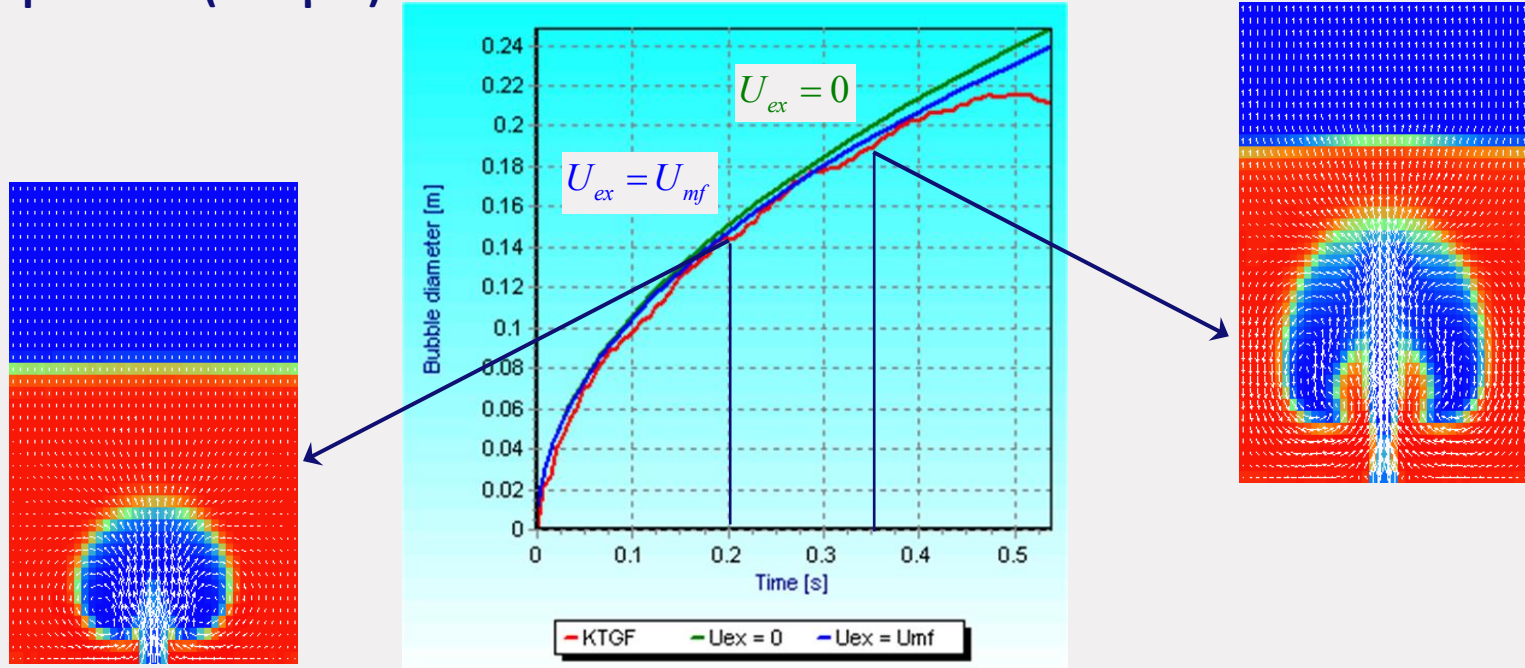
Simulation

Experiment



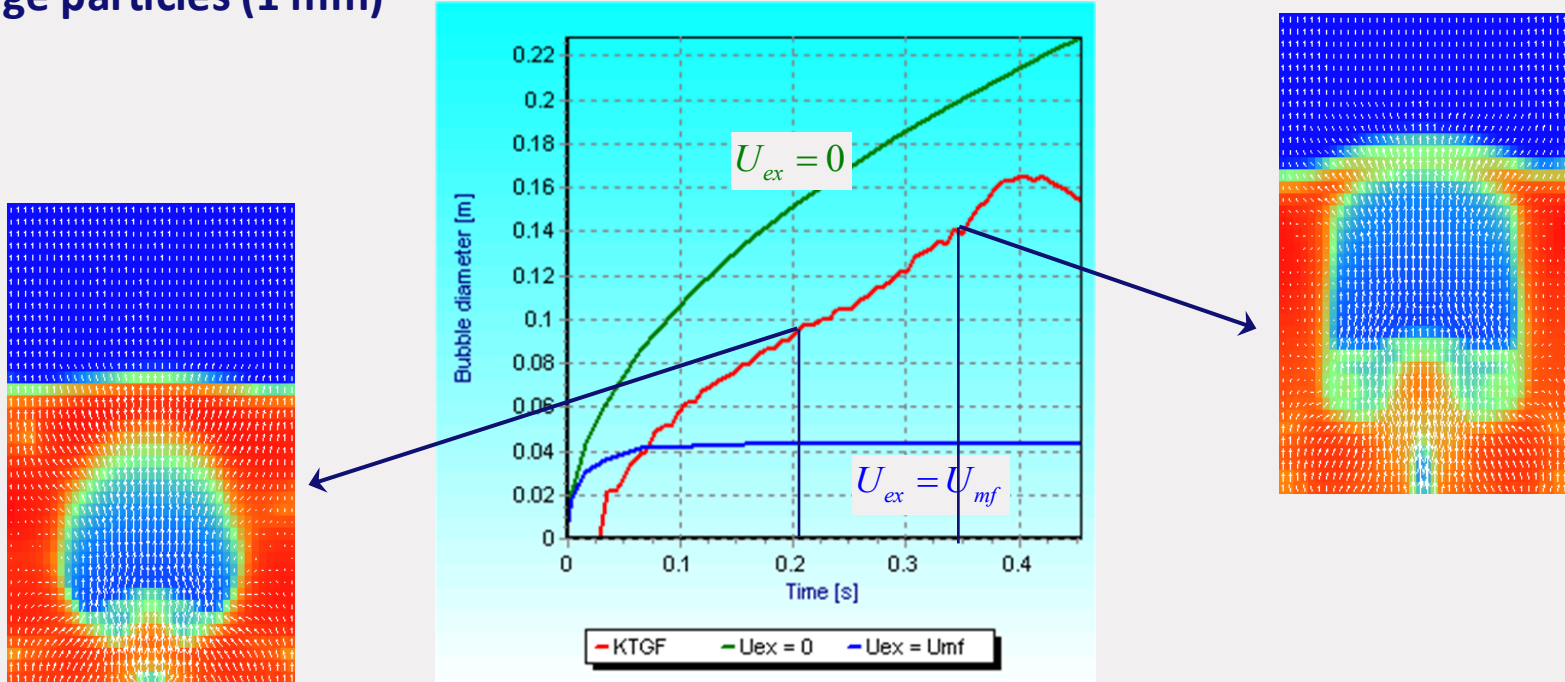
TFM: bubbling fluidized bed

Small particles ($100\text{ }\mu\text{m}$)

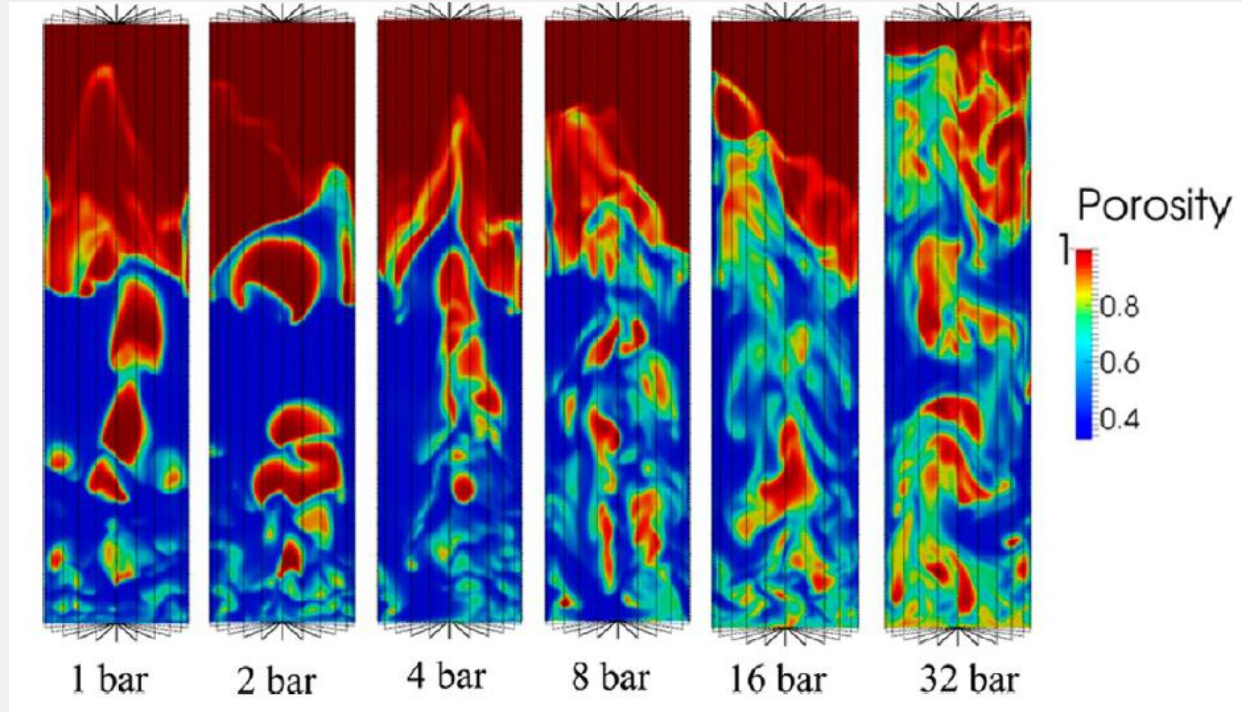


TFM: bubbling fluidized bed

Large particles (1 mm)



TFM: effect of pressure on bed dynamics



Commercial CFD packages

CFX, FLUENT, Star CCM+, COMSOL, etc.

Advantages

- Easy pre-processing (grid generation)
- Easy post-processing
- Multi-purpose (?)

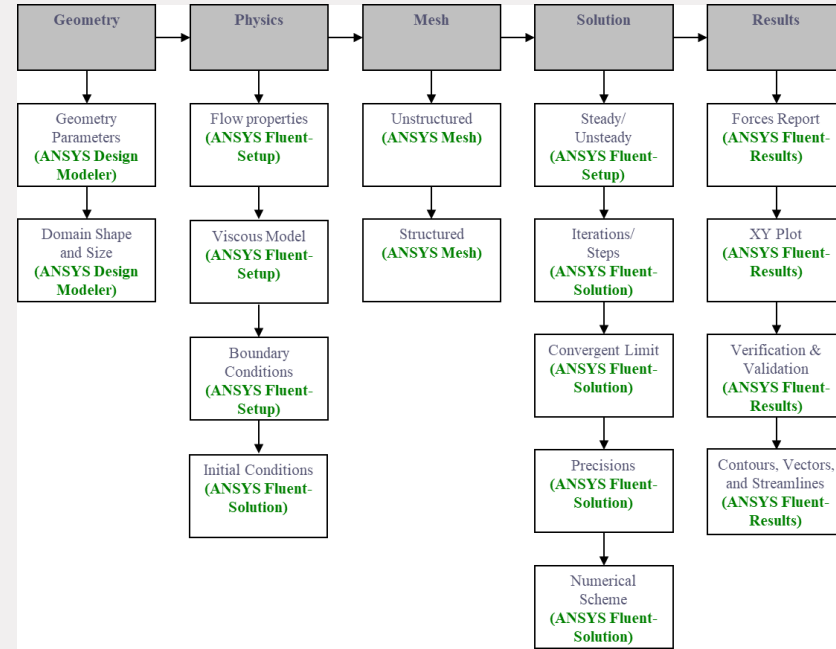
Disadvantages

- Difficult to include user models
- Flow solver is mostly a black box
- Solvers not optimized for flow problems

Commercial CFD packages

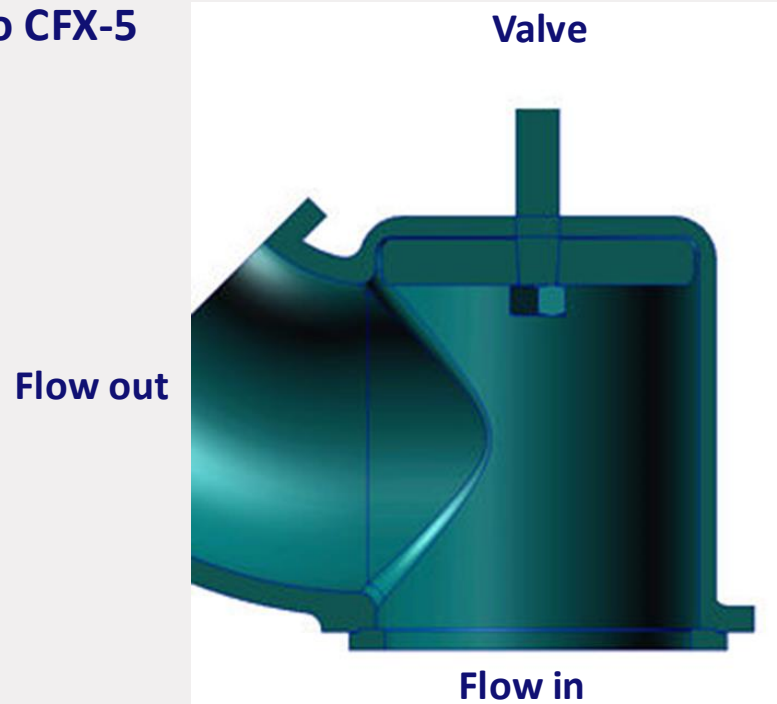
CFD process

- Geometry description
- Grid generation, transformation, refinement
- Specification of flow conditions and properties
- Selection of models
- Specification of initial and boundary conditions
- Specification of numerical parameters
- Flow solution
- Post processing: Analysis and visualization



Example: slurry control valve

- Step 1: Import CAD design data into CFX-5



Example: slurry control valve

- **Step 2: grid generation**

Automatic grid refinement at important locations

Structured grids (simple geometry) or

Unstructured grids (complex geometry)

more information see chapter 11



Example: slurry control valve

- Step 3: Solve the model

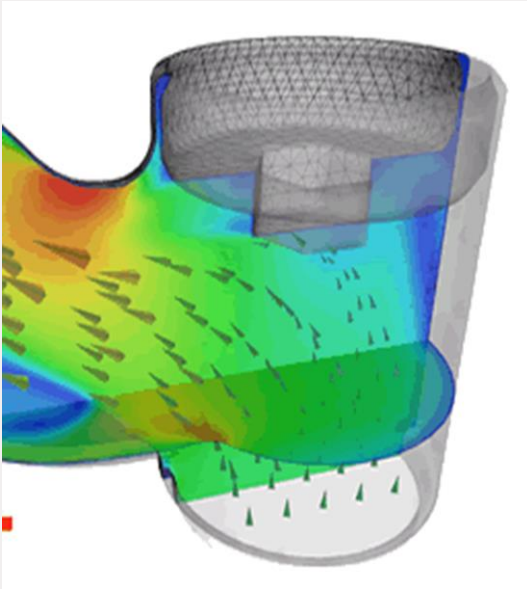
Problem-description in command language

```
>>CFX5
>>OPTIONS
  THREE DIMENSIONS
END
>>MODEL DATA
  >>PHYSICAL PROPERTIES
    >>FLUID PARAMETERS
      VISCOSITY 1.0E-3
      DENSITY 1.0E3
    END
  >>MODEL BOUNDARY CONDITIONS
    >>INLET BOUNDARIES
      PATCH NAME 'INLET'
      U VELOCITY 0.707
      V VELOCITY 0.707
      W VELOCITY 0.0
    END
    >>PRESSURE BOUNDARIES
      PATCH NAME 'PRESSURE BOUNDARY'
      PRESSURE 0.0
    END
  >>STOP
```

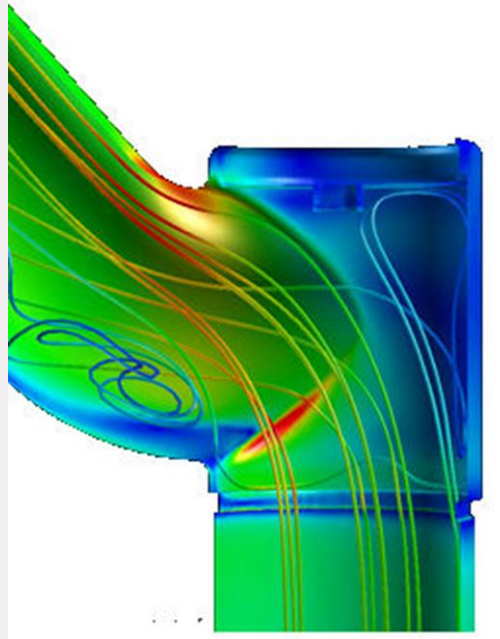
Example: slurry control valve

- Step 4: post-processing

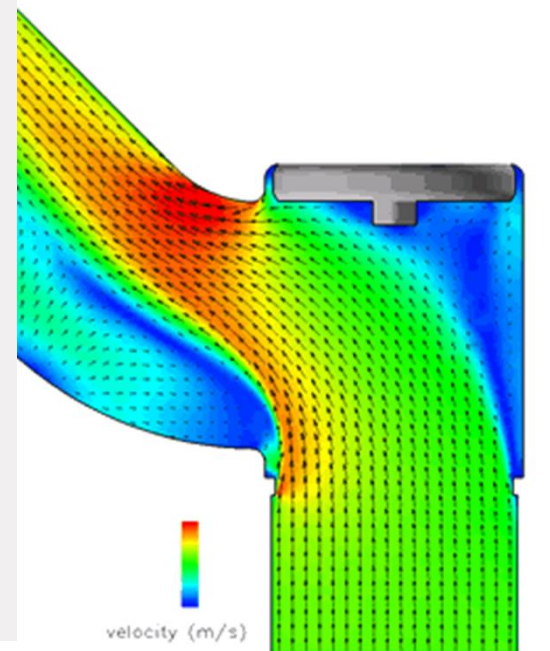
3D view



2D view (streamlines)



2D view (flow field)



More information

CFX	<u>https://www.ansys.com/products/fluids/ansys-cfx</u>
FLUENT	<u>https://www.ansys.com/products/fluids/ansys-fluent</u>
Star CCM+	<u>https://mdx.plm.automation.siemens.com/star-ccm-plus</u>
COMSOL	<u>https://www.comsol.nl/</u>

Open-sourced CFD codes

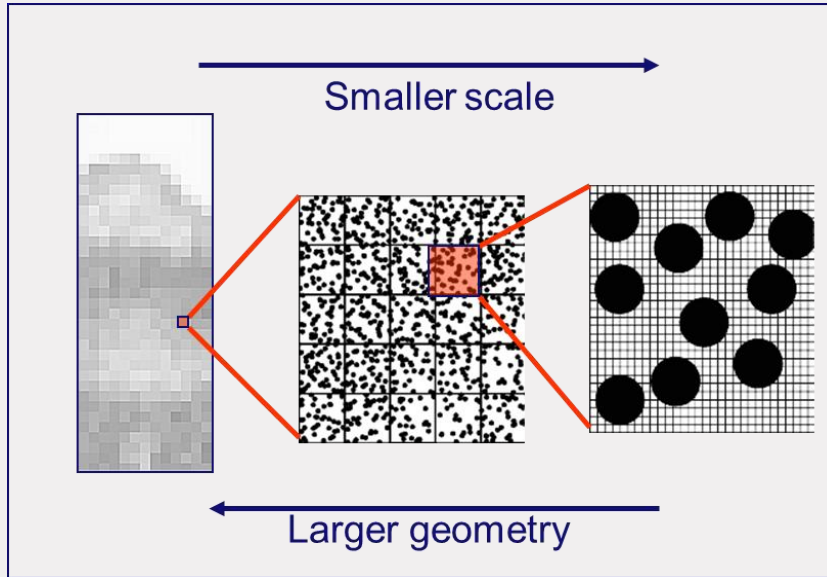
OpenFOAM	<u>https://www.openfoam.com/</u>
CFDEM	<u>https://www.cfdem.com/</u>

- News, forums, wiki, links, jobs, books, events

CFD-online	<u>https://www.cfd-online.com/</u>
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Wrap up

- **Multiphase flows: multiscale modelling**



- **Commercial CFD packages**

Easy to use

Good knowledge of CFD is still necessary

- **Open-sourced codes**

Large user community